

4(a)	progressive waves transfer energy or stationary waves do not transfer energy	B1
4(b)(i)	0.32 m	A1
4(b)(ii)	$v = \lambda / T$ or $v = f\lambda$ and $f = 1 / T$	C1
	$v = 0.32 / 0.020$ or 50×0.32 $= 16 \text{ m s}^{-1}$	A1
4(b)(iii)	450° or 90°	A1
4(b)(iv)	(P has) maximum downward displacement at 0.005 s	B1
	returns to original position/point (at 0.010 s)	B1
4(c)(i)	(position where) zero amplitude	B1
4(c)(ii)	2	A1
4(c)(iii)	180°	A1
4(c)(iv)	string drawn between X and Y with one antinode midway along the string	B1